Inferring State Technical Capacity and Preferences from ABAWD Waiver Allocations *

Julian Duggan and Josh Levy

June 3, 2026

PRELIMINARY AND INCOMPLETE; DO NOT SHARE

Abstract

Since its founding, the Supplemental Nutritional Assistance Program (SNAP) has included work requirements for select recipients. Since the Personal Responsibility and Work Opportunity Reconciliation Act of 1996, these requirements have been especially strict for Able-Bodied Adults Without Dependents (ABAWDs). However, state governments could apply for waivers from these ABAWD work requirements in areas of their state that met a complex set of conditions designed by the US Department of Agriculture's Food and Nutrition Services (USDA FNS). The recent release of FNS application archives provides a unique opportunity to study how states navigated this complex decisionmaking environment. We study whether state government approached these applicaitons with a view towards maximizing the number of ABAWDs waived from the requirements. In those cases where state applications do not exhibit ABAWD-waiver-maximizing behavior, we develop tests to indicate whether technical capacity constraints or policymaker preferences better explain the sub-optimal behavior. In a case study of North Dakota, we find evidence ABAWD-waiver-maximizing behavior limited by technical constraints in earlier years. In a case study of Wisconsin, we find no evidence of capacity constraints but are unable to distinguish between the theories of state preferences.

*We are particularly appreciative of Reid Kelley at the Bureau of Labor Statistics for explaining, collating, and sharing the vintages of the Local Area Unemployment Statistics.

1 Background

1.1 SNAP and ABAWD Work Requirements

The Supplemental Nutritional Assitance Program (formerly the Food Stamps Program) has been a core component of the US welfare system since the mid-1970's, and it currently disburses approximately $100 billion dollars a year to roughly one in eight Americans. Since its founding, the program has included general work requirements that mandate many recipients register for work, accept jobs if offered, maintain at least 30 hours of work per week if employed, or participate in Employment and Training programs. Beginning with the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA), Able-Bodied Adults Without Dependents (ABAWDs) were subject to additional work requirements in order to receive food stamps for more than 3 months during any 36 month period. These mandated that recipients work or participate in work and training programs for at least 20 hours per week. Importantly, time spent searching for work does not count towards these 20 hours. While general work requirements cover about 35% of SNAP recipients, ABAWD requirements cover just 9% [Bauer and East, 2026].

Lawmakers acknowledged that ABAWDs living in areas with high unemployment might fail to meet these requirements despite their best efforts to find work. They added a provision to the bill that permitted state governments to apply for exemptions from these requirements for ABAWDs living in areas of their state that met either of two criteria:

[**10% Rule**] Has an unemployment rate of over 10 percent [1]

[**Insufficient Jobs**] Does not have a sufficient number of jobs to provide employment for the individuals.

[1] TO DO: Check how many applications are approved under this rule rather than the insufficient jobs criterion. On the one hand, states can self-certify waivers under this rule and begin isssing them at the time the application is submitted, rather than waiting until it is approved. So they have timing-based incentives to use this criterion. On the other hand, barring these timing issues, at almost all points in time, any area that can qualify under this rule may also qualify under the [20% Rule] stated below. And invoking the [20% Rule] expands the set of other areas that can waived due to grouping. Therefore, we want to argue that invoking this rule is usually weakly dominated for most applications.

The US Department of Agriculture's Food and Nutrition Services (USDA FNS) was charged with operationalizing these criteria, especially [Insufficient Jobs]. Their initial guidance was published in 1996, with updates published in 2006, 2016, and 2021. These guidance documents established a complex decision environment for state policymakers interested in obtaining these waviers. In what follows, we discuss the core aspects of the approval process, but a more complete discussion of this process can be found by reading these guidance documents themselves [U.S. Department of Agriculture, Food and Nutrition Service, 1996, 2006, 2016, 2021].

1.2 Research Questions

The recent release of FNS application archives provides a unique opportunity to study how states navigated this complex decisionmaking environment. It allows us to ask the following interrelated questions:

1. Did state policymakers navigate this process "optimally" i.e. in a way that maximized the number of ABAWDs receiving waivers?

2. To the extent that policymakers didn't optimize, is this failure better explained by a lack of technical capacity or by their underlying preferences?

3. Which account of their technical constraints best explains deviation from optimal behavior? For instance, did policymakers use simple, sub-optimal heuristics to make computationally complex decisions? Did they forgo some degrees of freedom permitted by the application process? Which and why?

4. Which account of their preferences best explains devition from optimal behavior? For instance, policymakers may have a preference for the imposition of work requirements or show favoritism to some areas of their states over others.

5. How much of the gap from optimal behavior can be explained by policymaker observables e.g. party affiliation, technical background, experience, election-proximity? Are choices by policymakers of different types better explained by the technical capacity or preference accounts?

6. Depending on our answers to the above questions, we may find that the complexity of the application process created exogenous variation in waiver allocations across sub-state areas. Can we use our sharpened understanding of the sources of this variation to improve on prior studies that use variation in ABAWD waiver allocations to study the impact of work requirements on SNAP participation and other outcomes downstream of SNAP [Han, 2022, Lippold and Levin, 2021]?

1.3 The Insufficient Jobs Criterion

In the original FNS guidance, an area qualifies for a waiver under the Insufficient Jobs Criterion if it:

[**LSA**] Is designated as a Labor Surplus Area (LSA) by the Department of Labor's Employment and Training Administration (DOL ETA);

[**EB**] Is determined by the DOL ETA's Office of Unemployment Insurance (OUI) Service as qualifying for extended unemployment benefits;

The ETA gives LSA designations to civil jurisdictions with unemployment rates at least 20% higher than the national unemployment rate [2] [3] Similarly, the OUI issues an [EB] trigger for

[2] The designation rule also has ceiling and floor provisions that override the 20% rule. No jurisdiction unemployment rates below 6% can be designated LSAs, while jurisdictions with unemployment rates above 10% are automatically designated LSAs.

[3] Civil jursdiction include the following kinds of areas:

- A city of at least 25,000 population on the basis of the most recently available estimates from the Bureau of the Census; or
- A town or township in the States of Michigan, New Jersey, New York, or Pennsylvania of 25,000 or more population and which possess powers and functions similar to those of cities; or
- A county, except those counties which contain any type of civil jurisdictions defined in A or B above;

any state that reaches threshold levels for its 13-Week Insured Unemployment Rate (IUR) or 3-Month Total Unemployment Rate (TUR). The methods used to calculate these figures are nuanced; see [Corson and Rangarajan, 1993] for a more detailed discussion. For our purposes, the essential point is that FNS effectively chose numerical thresholds and binary DOL designations to operationalize the Insufficient Jobs Criterion.

FNS also included a handful of softer criteria. They indicated that waivers may be available to areas experiencing a "lack of jobs due to lagging job growth" or a "lack of jobs in declining occupations or industries." These criteria were softer in the sense that they were not accompanied by necessary or sufficient quantitative thresholds and administrative categorizations. Later versions of the guidance substantially reduced the discussion of these softer criteria, and in practice, few waivers were ever approved on their basis alone.[4]

The 1996 guidance set several other important precedents. First, it recommended that states cite official Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) figures when justifying a waiver in terms of the area's unemployment rate. Because LAUS figures are generated for a pre-specified set of geographies, a norm was established that waivers were typically granted to the LAUS geographic units: states, counties, cities with a population of at least 25,000, cities and towns in New England, and select metropolitan and small labor market areas. [5]

Second, it explicitly allowed states to combine geographic units (e.g. counties) into groups when calculating unemployment rates:

> *USDA will give States broad discretion in defining areas that best reflect the labor market prospects of program participants and State administrative needs...*

---

or

- A "balance of county" consisting of a county less any component cities and townships identified in paragraphs A or B above; or
- A county equivalent which is a town in the States of Connecticut, Massachusetts, and Rhode Island, or a municipio in the Commonwealth of Puerto Rico.

[4]TO DO: Confirm number of applications approved on this basis in the data.

[5]TO DO: What about the stranger of LSA's civil jurisdictions? How often were waivers given to the "balance of a county"?

> *Accordingly, states should consider areas within, or combinations of, counties, cities, and towns...*

The 2016 guidance clarified that these groups should be either geographically contiguous or considered part of an "economic region," though the approval data indicate the contiguity restriction was respected in (nearly?) every approved application since the beginning. [6]

Finally, the 1996 guidance emphasized that waivers were typically to be granted for one year at a time.[7]

In 2006, the guidance included a third sufficient criteria:

[**20% Rule**] Has a 24-month average unemployment rate 20% above the national unemployment rate for the same 24-month period.

This new criteria amounted to a extension of the LSA designation criterion to the geographic units found in the BLS LAUS dataset, especially counties that contain cities. It was accompanied by a few other sharpened rules. First, the 2006 guidance mandated that unemployment calculations be done using BLS' non-seasonally-adjusted unemployment and labor force series.[8] Second, it gave a standardized aggregation and rounding procedure that left no ambiguity in how states were to use the BLS series to determine area unemployment rates or national thresholds when applying under the [20% Rule]. Third, it clarified that no geographic unit could be "double-counted" in an application by being included in more than one wavied group. Fourth, it restricted the windows of time during which an area's unemployment rate could justify a waiver. It specified that the 24-month window used to justify a [20% rule] exception could begin no earlier than the January of two fiscal years prior to the fiscal year when the requested waiver would take effect. This window selection rule

[6] TO DO: Confirm that discontinuous regions are exceptionally rare throughout the time series. I suspect the 2016 guidance was updated to reflect what was already a *de facto* rule.

[7] TO DO: Check how many waivers are granted for other durations. Based on language in the guidance, the most common ones to expect are three-month periods to cover seasonal unemployment and two year windows in places with 36 months of high unemployment.

[8] TO DO: Count the number of applications, both pre and post 2006, approved on the basis of data other than the non-seasonally-adjusted unemployment and labor force series. I suspect that post-2006, the answer for non-reservation areas is effectively 0. Prior, it may be higher, in which case our strategy is imperfect.

was taken directly from LSA designation rules. Fifth, it clarified that there was no limit to the number of times a state could apply for these waivers throughout the year. [9] In pratice, most states in most years continued to file waiver applications on a regular annual cycle.[10]

Further revisions to the guidance in 2016 and 2021 maintained these core features of the 1996 and 2006 guidance. In 2025, the One Big Beautiful Bill erased the Insufficient Jobs Criterion and banned area grouping. Today, the regulation requires an individual geographic unit to qualify on its own under the [10% Rule].[11]

1.4 Applicants' Degrees of Freedom

Reviewing these waiver criteria and related aspects of program history, we can identify several conditions under which the state's application process is very straightforward:

a. State-years that qualified under [EB] were automatically granted a year-long, statewide waiver.

b. From April 1, 2009 to September 30, 2010, the American Recovery and Reinvestment Act (ARRA) suspended the ABAWD time limits. The Families First Coronovirus Response Act (FFCRA) of 2020 did the same from April 1, 2020, to June 30, 2023. States did not need to apply for waivers during either of these periods. [12]

c. State-years where 0 BLS geographic units or LSA civil jurisdictions individually qualify in any 24-month window leave no remaining decisions for the state to make.

However, in state-years not covered by these exceptions, the eligibility criteria were complex. They left open several avenues through which motivated, technically sophisticated policy-makers could exert additional effort to increase coverage of their wavier. These included:

[9] By 2016, this last point was sharpened further to indicate that states could discontinue or amend existing waivers on an ongoing basis as they pleased.

[10] TO DO: Put a number on this.

[11] TO DO: Include reference to latest FNS guidance when it is issued. New guidance will be posted here.

[12] However, if they desired, states were permitted to impose the ABAWD time limits on individuals who refused qualifying workfare slots. We can use this behavior to identify state governments with preferences for work requirements.

I. Varying the window of BLS data used to justify a waiver. The [20% Rule] stipulates that states can qualify with any consecutive 24-month window beginning after the January of two fiscal years prior to the waiver start date. Depending on the application date and the requested waiver start date, this leaves states with many possible windows to choose from, each of which would have its own national threshold and area unemployment rates to compare.

II. Varying the application date and waiver start date. Conditional on their application month, a state could increase the number of windows available to choose from by 12 if their start date falls in or before September – part of the current fiscal year – rather than in or after October, the beginning of the subsequent fiscal year. By applying closer to the waiver start date, the number of windows available to choose from increases as BLS publishes more recent data.

III. Varying the set of qualifying geographic units. States that analyzed unemployment rates of different geographic units – cities, counties, and metropolitan areas – could expand their waiver coverage beyond what is possible when only considering one geographic unit type.

IV. Thoughtful aggregation of geographic units into areas. States that considered a variety of contiguous county grouping strategies could expand the scope of qualifying areas, while states following simple grouping heuristics might leave some waiver scope on the table.

V. Particularly in earlier years, testing the boundaries of these rules in a handful of other respects: using older vintages of the BLS data, using data sources other than the BLS, using BLS' seasonally adjusted data, using non-standard averaging practices, constructing non-contiguous geographic groups, relying on non-BLS geographic units, applying under the softer standards of "lack of jobs due to lagging job growth" or "lack of jobs in declining occupations and industries" [13]

In section 2, we share descriptives from the application data on each of these possible degrees of freedom. We show that item I. and item IV. are the main strategies explored by states to expand their waiver coverage.[14] Then, we formalize the state's application problem in years where the state decision problem was not bypassed by [EB] qualification, national requirement suspensions, or very low unemployment rates.

2 Data and Descriptives

2.1 Data

BLS LAUS vintages

- County-month[15] unadusted labor force and unemployment counts, based on XXX methodologies over time.
- Month t metropolitan new releases since before PRWORA coincide with preliminary BLS release of county-month figures for the month $t-3$, and revised county-month figures for the month $t-4$.
- Annual January data release (typically in March) coincides with revision of many years' past data.
- Currently, data are restricted to post-January 2011 vintages of post-January 2011 data. BLS appears not to have archived pre-2011 vintages or data. [16]

[13] TO DO: Put numbers on each of these.

[14] TO DO: criteria allowing us to make this assertion?

[15] And city/metropolitan area, if our preliminary results suggest these are a common geography. Implicitly, the word "county" encompasses the planning regions of Connecticut, boroughs of Alaska, parishes of Louisiana, ... So it should always be understood to mean "county-equivalent" throughout this document.

[16] TO DO: Find earlier vintages, or else develop clear arguments of what can be done in earlier years in the absence of these data. The prognosis is surprisingly bad for the county-month vintages. Three methods exist extending our county-month vintage data further back in time. One is to scrape the webarchive, which is very likely to get vintages back to early 2010. Earlier vintages are not available on webarchive because BLS

FNS application guidance from 1996, 2006, 2016, and 2021 report the calculation methods underpinning our analysis.

FNS waiver applications and responses:

- N state-years
- N applications
  - Observe full application, permitting more thorough analysis of V.
  - Observe FNS response only, where (we believe) a thorough analysis of all applicant degrees of freedom named in subsection 1.4 can be performed.[17]
- N applications where no data availability restrictions or simplifying conditions i.e. item a., item b., or item c., bypass state decision problem. This is our effective sample size for the core analysis as it currently stands. Call this the "main sample."[18]
- N county-applications in the main sample.

BLS LN (source of the national unemployment rate use to calculate the [20% Rule] thresholds. There ar no vintage considerations here), CPS (source of county valuations), SNAP participant projections (for July of each year. from FNS. not yet made all the way to 2026.

changed its data disemmination links over time. To get further back, much Codex searching has failed to turn up the full county-month data. A second approach is to use the ALFRED archive. These likely contain vintages back to 2007 at the county-month level. Here, we observe labor force counts but unemployment *rates* rounded to a single decimal point. Third, we could explore a comment from Reid Kelly indicating that revised county series are usually only reflecting revised state-level totals for unemployment and labor force counts, and we have state-level vintages of the data back to 2008. We can impute county shares of state totals in each vintage back this far. As a robustness check, we could then confirm that this approach produces (nearly) identical estimates to the county-level series in later years where we can observe both. This approach still will not take us prior to 2008, and it may be dominated by the ALFRED archive. In pre-2007 application years, we may use the earliest vintage we can find for each county-month, and develop an argument that changes were "as good as random" going back to earlier vintages.

[17]TO DO: Confirm this statement or qualify it.

[18]TO DO: Add additional conditions under which an application is disqualified from our main sample e.g. if acceptance norms were wildly different pre-2006 guidance, so that our model of the state application problem is too wrong to be credible.

may be dominated by doing our own calculations in the CPS.), county adjacency matrices, ...

2.2 Descriptives

Check each of the TO DO's in the footnotes above for tasks to be done in this section. A main task of the descriptives is to explore how applicants used each of the degrees of freedom noted above. To that and related ends, we should produce time series of each of the following. It may often be useful to disaggregate these time series by policymaker observables e.g. party affiliation, technical background, tenure length, election cycle timing...

2.2.1 State-Year-Level

- N state-years qualifying under item a., item b., and item c.
- Average share of each year with gaps in waiver coverage. We expect this to be quite low and/or trending down over time. If it is not, disaggregating these shares by state observables may speak to some of our research questions

2.2.2 Application-Level

- N applications lost to vintage data limitations (or imputed with closest available vintage)
- N main sample applications
- Share main sample applications invoking each qualification rule.
- Share sample applications using a window other than the most recent available or first permissible, speaking to item I..
- Among main sample applications (or across all applications? perhaps we don't need to restrict to the main sample here), average months before October of waiver start date,

speaking to item II.. Applicant learning is consistent with this chart trending down over time.

- Among main sample applications (or across all applications? perhaps we don't need to restrict to the main sample here), average months' gap between application submission and waiver start, speaking to item II.. Applicant learning is consistent with this chart trending down over time.
- Share main sample applications using geographies other than counties and multiple geography types, speaking to item III.. Applicant learning is consistent with this chart trending up over time.
- Share main sample applications constructing groups, especially groups that include individually ineligible counties, speaking to item IV.. Applicant learning is consistent with this chart trending up over time.
- Share main sample applications creating multiple, distinct groups, speaking to item IV.. Applicant learning is consistent with this chart trending up over time.
- Share main sample applications usings different windows of data to justify waivers for different groups, speaking to item I. and item IV.. Applicant learning is consistent with this chart trending up over time.
- Share of main sample applications exploring any other miscellaneous methods of rule bending as found in item V..

2.2.3 County-Application-Level

- N county-applications in the main sample where the county qualifies on its own under the [10% Rule] or [20% Rule]
- N county-applications where the county is added as a group member under the [10% Rule] or [20% Rule], but does not qualify on its own terms.

3 Model

We now present a model of the state application process that underpins our measurement approach. State s applies at time t^{app} for a one year waiver that begins at t^{start}. Counties in state s are indexed by $i = 1, \dots, I$. By fixing the application and waiver start time, we are assuming that some exogenous process is determining the application timing, or that states are already optimizing their timing choices. We also assume away the several geographic levels that can qualify – especially cities and metropolitan areas – and assume counties are the only permissible geographic level. Further work can improve on this simplification.

3.1 Fixed-Window Problem

Fix a particular data 24-month window $w \in W(t^{app}, t^{start})$. We first suppose the state can apply using data from this window alone, but we will generalize the problem to account for multiple window options in the next subsection. For each $i \in s$, we observe window-level data:[19]

$$\{L_i, U_i, V_i\}_{i=1}^{I}$$

where $L_i := \sum_{t \in w} L_{it}$ denotes the 24-month labor force, $U_i = \sum_{t \in w} U_{it}$ denotes the 24-month unemployment count, and V_i denotes the policymaker's subjective value of receiving an ABAWD waiver for county i. [20] Fix the national unemployment rate $\tilde{\alpha}$ during w so that the unemployment threshold is given by $\min\{1.2 * \tilde{\alpha}, .1\} := \alpha \in [0, .1]$. The min operater incorporates the [10% Rule] encoded in PRWORA. The policymaker applies to waive ABAWD requirements in a collection of m county groups $g_1, \dots, g_m \subseteq \{1, \dots, I\}$ which are pairwise disjoint, individually geographically contiguous, and individually satisfy

[19] We suppress all state, time, and window subscripts as these are held fixed.

[20] Value could be measured by ABAWD counts, by county counts (so that $V_i = 1$ for all counties), or by some other objective of the policymaker.

the unemployment threshold condition:

$$\frac{\sum_{i \in g_j} U_i}{\sum_{i \in g_j} L_i} \geq \alpha \quad \text{for each } j = 1, \ldots, m$$

We can state this unemployment condition in terms of county-level "slack" $s_i := U_i - \alpha L_i$:

$$\sum_{i \in g_j} s_i \geq 0 \quad \text{for each } j = 1, \ldots, m$$

Formally, policymaker's objective is to choose a set of groups of counties in order to maximize total value of waived areas: [21]

$$\max_{G \in \mathcal{P}(\mathcal{P}(I))} \sum_{g \in G} \sum_{i \in g} V_i$$

subject to

$$g_j \text{ spatially contiguous } \forall g_j \in G \tag{1}$$

$$g_j, g_{j'} \in G \implies g_j \cap g_{j'} = \emptyset \tag{2}$$

$$\sum_{i \in g_j} s_i \geq \alpha \ \forall g_j \in G \tag{3}$$

Given solution $G^*_{swt^{app}t^{start}}$, the policymaker obtains their objective value

$$V^*_{swt^{app}t^{start}} := \sum_{g \in G^*_{swt^{app}t^{start}}} \sum_{i \in g} V_i$$

3.2 Cross-Window Problem

The problem is very similar in the case that the applicant has many windows of data to choose from. We introduce window subscripts where appropriate, but s, t^{app}, and t^{start} are still considered fixed. Given waiver valuations, county data, and national thresholds across

[21] $\mathcal{P}$ denotes the power set

all permissible windows:

$$\{\{L_{iw}, U_{iw}\}_{w \in W(t^{app}, t^{start})}, V_i\}_{i=1}^{I}$$

$$\{\alpha_w\}_{w \in W(t^{app}, t^{start})}$$

The policymaker maximizes their objective by choice of a set of groups of counties, where the groups are pairwise disjoint, individually geographically contiguous, and individually satisfy the unemployment threshold condition for some permissible window:

$$\max_{G \in \mathcal{P}(\mathcal{P}(I))} \sum_{g \in G} \sum_{i \in g} V_i$$

subject to

$$g_j \text{ spatially contiguous } \forall g_j \in G \tag{4}$$

$$g_j, g_{j'} \in G \implies g_j \cap g_{j'} = \emptyset \tag{5}$$

$$\forall g_j \in G,\ \exists w \in W(t^{app}, t^{start}) \text{ s.t. } \sum_{i \in g_j} s_{iw} \geq \alpha_w \tag{6}$$

Given solution $G^*_{st^{app}t^{start}}$, the policymaker obtains their objective value

$$V^*_{st^{app}t^{start}} := \sum_{g \in G^*_{swt^{app}t^{start}}} \sum_{i \in g} V_i$$

[22]

[22]**Timing Problem** We include a few comments in this section as a theoretical aside, but it may prove practically useful depending on how the project develops. Given policymaker in s at time t, they can observe $V^*_{stt^{start}}$ for each t^{start} before their current waiver expires, supposing static valuations. Presumably, a waiver start date that comes after the end date of the current waiver is very costly. For any fixed t^{start}, they can project the value of $V^*_{st't^{start}}$ for any $t' > t$. Their choice of t^{start} has dynamic implications for the next year's range of t^{start}.

3.3 Solution Strategy

Suppose there are $x < I$ counties with weakly positive slack. A natural solution strategy is given by this two-stage procedure.

First, identify the set of all admissible county groups. In the fixed-window setting, an admissible group is any subset of counties that satisfies the continuity condition and the slack condition. Every such group must contain at least one weakly positive-slack county, since a group made up entirely of negative-slack counties cannot satisfy $\sum_{i \in G_g} s_i \geq 0$. In the cross-window case, the continuity condition is unchanged while the slack condition can be satisfied in any one of the admissible windows. Let

$$\mathcal{C} = \{g_1, \ldots, g_K\}$$

denote the collection of all admissible county groups.

Second, solve a weighted set packing problem over the set of admissible groups. Formally,

$$\max_{y_1, \ldots, y_K} \sum_{k=1}^{K} W_k y_k$$

subject to

$$W_k = \sum_{i \in g_k} V_i. \tag{7}$$

$$\sum_{k:\, i \in g_k} y_k \leq 1 \text{ for each } i = 1, \ldots, I, \tag{8}$$

$$y_k \in \{0, 1\} \text{ for each } k = 1, \ldots, K \tag{9}$$

The solution to this weighted set packing problem $y^* := \{y_1^*, \ldots, y_K^*\}$ is a K-dimensional vector of binaries indicating which admissible groups should be selected by the policymaker. The problem amounts to a special case of a binary integer programming problem, which is NP-complete. The computational details by which this problem is solved are in the

Appendix.

4 Empirical Strategy

Our empirical strategy is tightly linked to the model. Again fix s, t^{app}, and t^{start}. The model derives optimal allocations under the assumption that policymaker valuations $\{V_i\}_{i=1}^{I}$ are known constants. In practice, valuations are unobserved. In what follows, we propose a method for inferring these valuations from observed choice data. Having inferred these valuations, we will explain how these inferences address our research quesitons.

4.1 Valuation Ratios R^j

Given any valuation scheme $V^j \in \mathbb{R}_+^I$, the model illustrated how we may solve for the policymaker's optimal county grouping strategy G^* and their maximized payoff $V^{j,*}$. In the FNS application data, we observe G^{obs}, the set of county groups actually waived. We can therefore calculate the observed payoff under valuation scheme V^j, namely:

$$V^{j,obs} := \sum_{g \in G^{obs}} \sum_{i \in g} V_i^j$$

Given valuation scheme V^j, we can ask how well it fits observed choices. To answer this question, we construct a valuation ratio:

$$R^j := \frac{V^{j,obs}}{V^{j,*}}$$

Notice that $R^j \in [0,1]$ as the payoff of the observed waiver allocation can never be higher than the payoff of the optimal waiver allocation under V^j. The closer R^j lies to 1, the closer the observed choice lies to the $j-$optimal choice in terms of V^j. Given two different valuation schemes V^j and V^k, we can identify which better explains the observed allocation by comparing their R values. V^j is a better fit with observed choice data whenever $R^j > R^k$.[23]

4.2 Addressing the Research Questions

We consider three different valuation schemes that could have plausibly guided policymaker decisionmaking. The first is a county count scheme, where $V_i^N = 1$ for all i. A decisionmaker allocating waivers according to this scheme aims to maximize the number of counties that are wavied, regardless of their other characteristics. The second is a population scheme, where V_i^{pop} is equal to the size of the population living in i in the application year. A decisionmaker allocating waivers according to this scheme aims to maximize the number of people living in waived counties. The third is a SNAP population scheme, where V_i^{SNAP} is equal to the number of SNAP recipients living in i in July of the application year. A decisionmaker allocating waivers according to this scheme aims to maximize the number of SNAP recipients living in waived counties.

Under the assumption that a policymaker wants to waive the requirement for as many SNAP recipients as possible, we can interpret these three schemes, V_i^N, V_i^{pop}, and V_i^{SNAP} as reflecting increasingly levels of policymaker sophistication. The least sophisticated policymaker aims to maximize the number of waived counties. The most sophisticated aims to maximize coverage of the SNAP population. [24]

[23] Beyond this ordinal interpretation, the meaning of R^j is nebulous. It captures the share of total possible V^j value realized by the observed choice, but this share may not reflect all aspects of V^j-relative decision quality that are of interest to us. To develop more clarity, let $R^j(G)$ denote the valuation ratio under V^j when groups G are waived. This value has a distribution across the set of permissible G. Given one set of problem primitives, R^j's distribution across the $[.8, 1]$ interval may be smooth so that R^j at .8 indicates that many V^j-superior groupings were possible and missed by the decisionmaker. Given another set of problem primitives, R^j's distribution could be lumpy and full of gaps so that R^j at .8 indicates that very few V^j-superior groupings existed for the policymaker to select from. If we want to ask, "How many V^j-superior groupings were possible but missed by the decision maker?", then R^j alone will not tell us. Instead, we may want some to measure where $R^j(G^{obs})$ lies in the distribution $R^j(G)$. This distribution will be computationally challenging to construct, so we must think carefully about whether/how to construct it.

[24] TO DO: Develop clearer arguments about the "right" valuation scheme. Consider the most sophsiticated, benevolent policymaker who doesn't care at all whether people work but only cares about total SNAP enrollment figures (this may not be benevolent objectively, but it's a baseline – should they also care about disbursement amounts? total income in the population?). What do they target? They consider potential enrollment outcomes for each county with and without the waiver, then choose the waiver allocation that maximizes total enrollment. Supposing SUTVA across counties, this is equivalent to targeting $V_i^{\tau} := Y_i(1) -$

We now describe our (tentative) approaches to addressing each of the research questions named above.

1. Did state policymakers navigate this process "optimally" i.e. in a way that maximized the number of ABAWDs receiving waivers?

   - We can address this question by looking at the level of R_{sy}^{SNAP} across states s applying in years y. If that number is below 1, we can estimate the number of SNAP-recipient-months (and, with additional assumptions, ABAWDs), who could have but did not receive the waivers.

2. To the extent that policymakers didn't optimize, is this failure better explained by a lack of technical capacity or by their underlying preferences?

   - We can address this question by offering accounts of what patterns technical capacity constraints generate in the data and what patterns non-ABAWD-waiver-maximizing preferences look like in the data. These patterns should be mutually exclusive so that we can sort each application into a single bucket based on is patterns. We then decide which buckets dominate in the panel. We'll specify these accounts and patterns in the answers to the quesitons below.

3. Which accounts of their technical constraints best explains deviation from optimal behavior? For instance, did policymakers use simple, sub-optimal heuristics to make computationally complex decisions? Did they forgo some degrees of freedom permitted by the application process? Which and why?

   - We give here a flavor of the kind of categorizations we could attempt. Whether these are sensible depends heavily on what we find in the data.

$Y_i(0)$, where Y_i is the number of enrolled ABAWDs without any waiver. If the poulation of ABAWDs in each i is homogenous in their treatment responses, then $V_i^\tau = \beta Y_i(0)$ and can be proxied for by $Y_i(0)$. What is the best predictor of $Y_i(0)$? Not observed, present-day Y_i since that is a function of the present waiver allocation. Probably a CPS measure of the likely number of ABAWDs in each county at the application date.

- An application with R^N or $R^{pop} = 1$ but $R^{SNAP} < 1$ is consistent with a simplifed decision procedure that targeted a poor proxy for the number of ABAWDs waived.
- An application with $R^{SNAP,w} = 1$ for the chosen 24-month window w but $R^{SNAP} < 1$ is consistent with partial window search.
- An application with $|G^{SNAP,*}| > 1$ but $|G^{obs}| = 1$, and G^{obs} optimal on the set $\{G \mid |G| = 1\}$ is consistent with single-group search.

4. Which account of their preferences best explains devition from optimal behavior? For instance, policymakers may have a preference for the imposition of work requirements or show favoritism to some areas of their states over others.

   - We give here a flavor of the kind of categorizations we could attempt. Whether these are sensible depends heavily on what we find in the data.
   - An application with $R^{SNAP} << 1$ and $|\{V^j - \text{superior, admissible } G\}|$ large is consistent with no attempt at V^{SNAP} optimization.
   - An application with $R^{SNAP} << 1$ and $R^X = 1$ where X is some county characteristic (e.g. population by race, population by party affiliation, population by income, etc.) is consistent with preferences for X over $SNAP$.
   - Let $V_i^{10\%} = \mathbb{I}\{unemp_{iw} \geq .1 \text{ for some } w \in W(t^{app}, t^{start})\}$. An application with $R^{SNAP} << 1$ and $R^{10\%} \approxeq 1$ and $|\{V^{10\%} - \text{superior, admissible } G\}|$ small, is consistent with preferences for work requirements, except in those places with unemployment rates where the original text of PRWORA guarantees the waiver.

5. How much of the gap from optimal behavior can be explained by policymaker observables e.g. party affiliation, technical background, experience, election-proximity? Are choices by policymakers of different types better explained by the technical capacity or preference accounts?

- Correlate bucket membership with these observables. Strong patterns would be required here to be compelling.

6. Depending on our answers to the above questions, we may find that the complexity of the application process created exogenous variation in waiver allocations across sub-state areas. Can we use our sharpened understanding of the sources of this variation to improve on prior studies that use variation in ABAWD waiver allocations to study the impact of work requirements on SNAP participation and other outcomes downstream of SNAP [Han, 2022, Lippold and Levin, 2021]?

    - TBA

5 Preliminary Results

We now present preliminary results for North Dakota and Wisconsin. The North Dakota case study presents a more interesting situation and supports the validity of our basic research design. The Wisconsin case study illustrates where our current data limitations, and the underlying variation in the data, limit the reach of our analysis.

5.1 North Dakota

Figure 1 reports valuations of the observed allocations and the cross-window optimal allocations according to the three valuation schemes V_i^N, V_i^{pop}, and V_i^{SNAP}. The sample is the 9 waiver applications submitted by North Dakota between 2013 and 2023.

The results are encouraging. First, the optimal valuation and observed valuation correlate well. This result suggests that policymakers are at least indirectly targeting the measures we've supposed. Second, only in the earliest years of the period did North Dakota policymakers select an allocation that was optimal according to one of our three schemes. In most years, they left some value on the table, and there is scope for our research design above to explore whether technical capacity constraints or preferences better explain the gap.

Third, we find that the observed valuation almost always lies below the optimal valuation. This result is reassuring because it should be true mechanically if we have correctly specified the application problem. However, we see two imperfections.

First, in 2023, the observed valuations surpass our model's optima. This outcome is possible only if the policymakers did not in fact face one or more of the constraints currently incorporated into our modelling of their problem. Inspecting the data, we believe there is imprecision in our assumptions about which vintage of the BLS dataset were available to North Dakota policymakers at the time they submitted the 2023 application. Their application fell near the time of the release of the 2022 annual data revision, so they likely faced a set of county labor force and unemployment rates other than those we have supposed. The vintage that they observed may have been much more pesimissitic than the one we've assumed i.e. it reported higher unemployment rates, allowing them to allocate waivers more widely.

Second, we see an overall drop in decision quality across all three metrics in 2018. This drop may be real. However, given the vintage misspecification likely contaminating the 2023 estimates, we want to be sure that vintage misspecification cannot explain this dip. If the vintage that they observed was more optimistic than the one we've assumed i.e. it reported much lower unemployment rates, then they would have been able to distribute waivers less widely than we've assumed.

In future work, we will sharpen our methods for imputing data release dates so that mismatches of this kind are no longer possible.

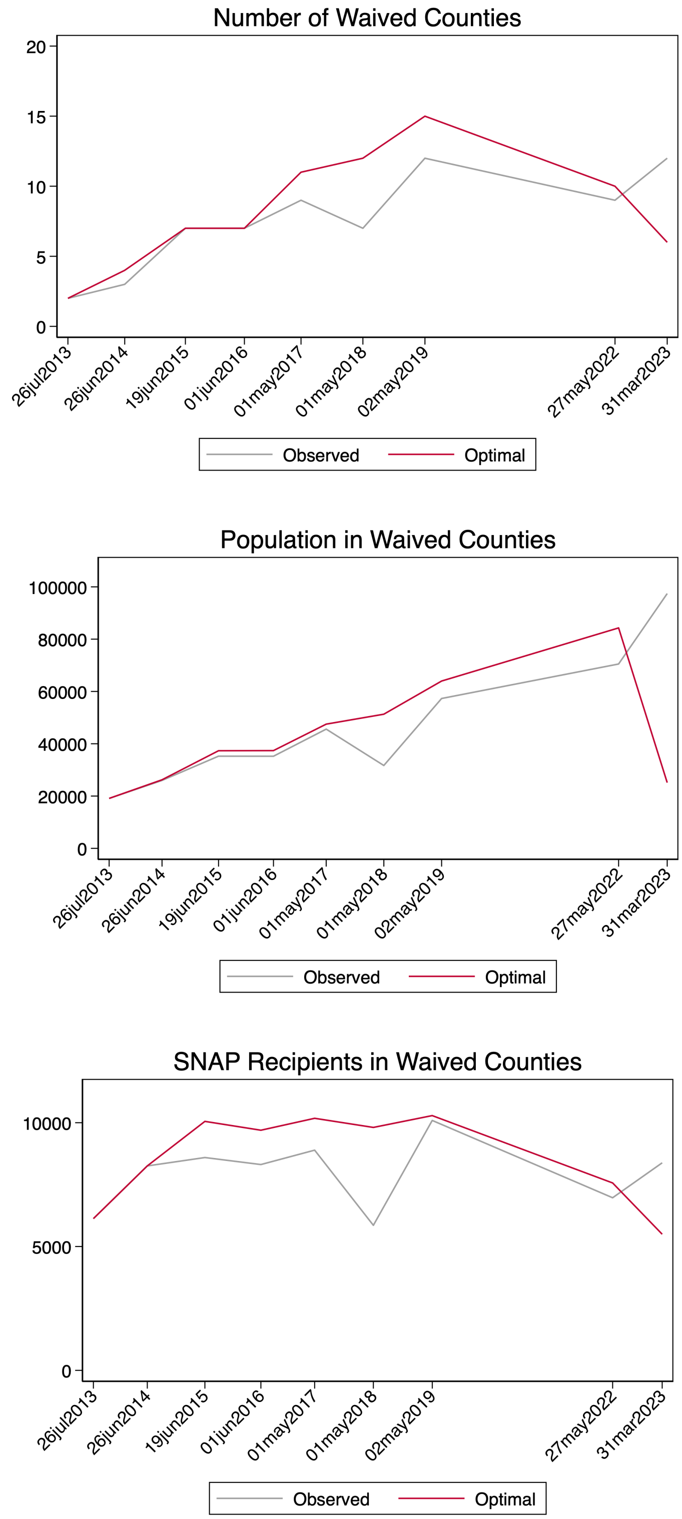
Figure 2 reports valuation ratios R^j according to each of the three valuation schemes using the cross-window optimum as the benchmark. In our model section, we argued that $R^j > R^k \implies V_j$ better explains decisionmaking than V^k.

Again, the results are encouraging. As our analysis above implied, the ratio is below one for every year except 2023. In the two earliest years, we see that two or more of the valuation schemes are optimized perfectly so that we cannot discriminate between them. In 2015 and 2016, the policymakers seemed to be optimizing for the number of waived counties, reflecting the least sophisticated approach according to our heirarchy. However, post-2018, the number of SNAP recipients is consistently the best predictor of their decisions, suggesting that they grew more sophisticated over time. In these later years where the unemployment data permit larger numbers of counties to be waived (as per the top panel of Figure 1), the dimension of their search problem was much larger. In the more complex decision environment, policymakers never found the optima. As before, this result is consistent with either a misspecified vaulation scheme or technical constraints.

5.2 Wisconsin

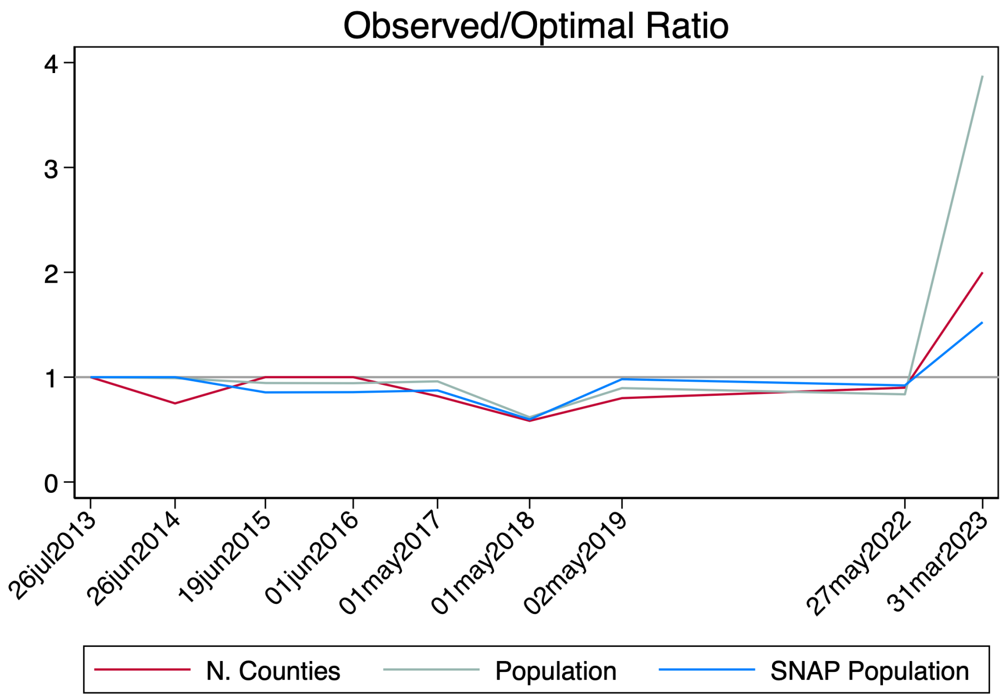
In Wisconsin, we find only two years of application data not censored by our data limitations. In each of these years, the size of the decision problem is small, and all three objectives we consider are optimized by the same group selection. So we cannot distinguish between theories state lawmakers' capacity and preferences.

Figure 1: North Dakota: Observed and Optimal Waiver Allocation Valuations



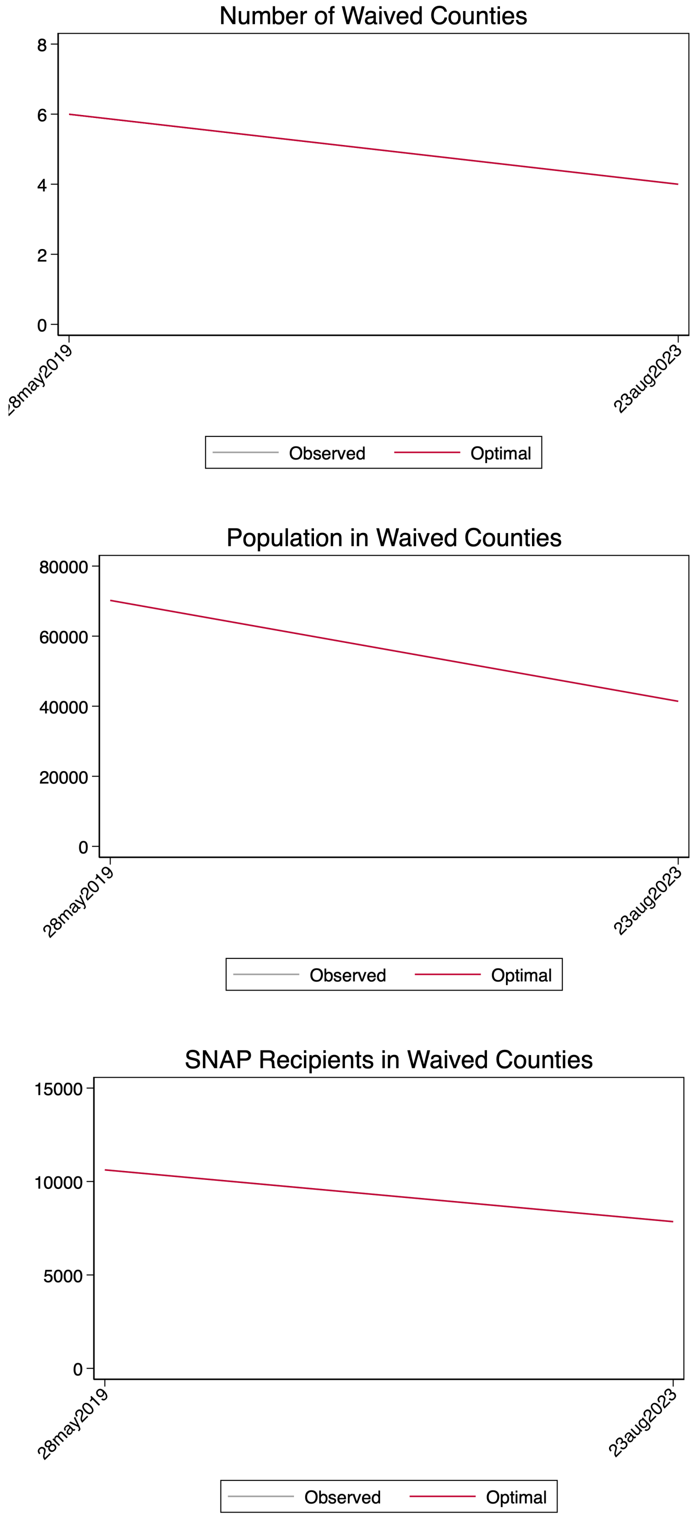
Notes: The panels report application values for North Dakota under county, population, and SNAP-population valuation schemes, using the cross-window optimum as the benchmark.

Figure 2: North Dakota: Valuation Ratios R^j



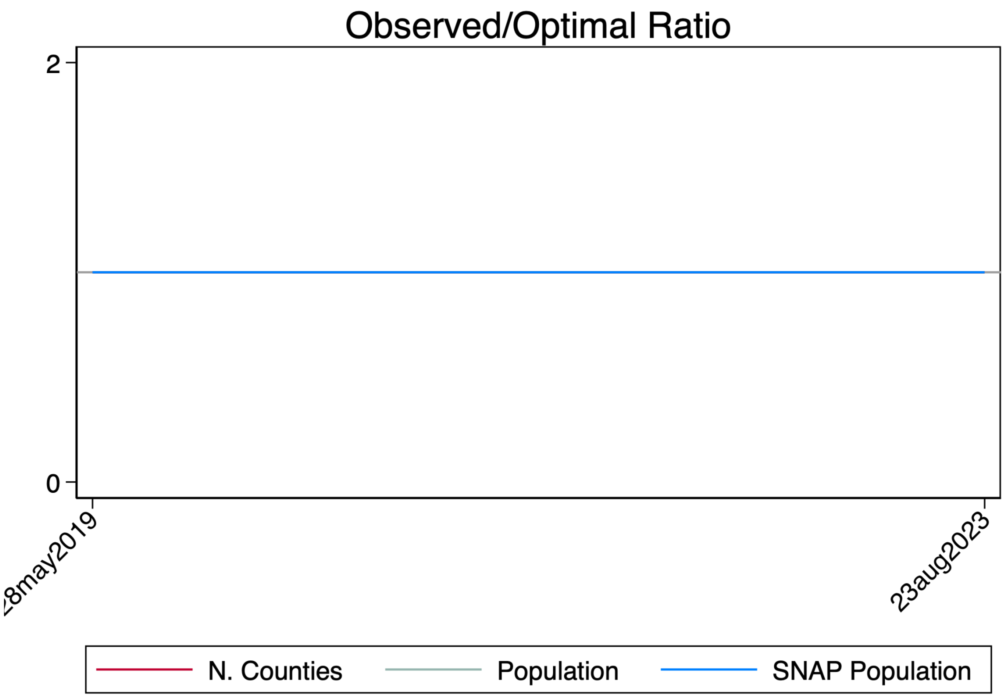
Notes: The figure reports valuation ratios for North Dakota's waiver applications using the cross-window optimum as the benchmark.

Figure 3: Wisconsin: Observed and Optimal Waiver Allocation Valuations



Notes: The panels report application values for Wisconsin under county, population, and SNAP-population valuation schemes, using the cross-window optimum as the benchmark.

Figure 4: Wisconsin: Valuation Ratios R^j



Notes: The figure reports valuation ratios for Wisconsin's waiver applications using the cross-window optimum as the benchmark.

References

Lauren Bauer and Chloe N. East. A primer on snap work requirements, 2026.

Walter Corson and Anu Rangarajan. Extended UI benefit triggers, 1993. Local PDF in literature folder: "Corson and Rangarajan (1993).pdf".

Jeehoon Han. The impact of snap work requirements on labor supply. *Labour Economics*, 74:102089, 2022.

Kye Lippold and Remy Levin. The effects of transfer programs on childless adults: Evidence from food stamps, 2021. SSRN working paper 3655794.

U.S. Department of Agriculture, Food and Nutrition Service. Guidance for states seeking waivers for food stamp limits, December 1996.

U.S. Department of Agriculture, Food and Nutrition Service. Guidance on requesting abawd waivers, August 2006.

U.S. Department of Agriculture, Food and Nutrition Service. Supplemental nutrition assistance program: Guide to supporting requests to waive the time limit for able-bodied adults without dependents (abawd), December 2016.

U.S. Department of Agriculture, Food and Nutrition Service. Supporting requests to waive the time limit for able-bodied adults without dependents, September 2021.

6 Appendix

6.1 Weighted Set Packing Solution: Computational Details

6.1.1 Identify the Set of All Admissible County Groups

Given (s, t^{app}, t^{start}), we first enumerate $\mathcal{C}$ for a particular window $w \in W(t^{app}, t^{start})$. The cross-window $\mathcal{C}$ will simply be the union of the window-specific $\mathcal{C}$ across all admissible windows. To enumerate $\mathcal{C}$ for a particular window, it is convenient to process positive-slack counties sequentially. Let the positive-slack counties be indexed so that

$$r_1 < r_2 < \cdots < r_x.$$

For each root county r_ℓ, we generate all admissible groups whose lowest-index positive-slack county is r_ℓ. This intermediate goal avoids duplication during search.

For a fixed root r_ℓ, the algorithm maintains an *active branch list*. Each branch is a pair (S, F), where S is the set of counties currently included in the candidate group and F is the set of counties that are forbidden from entering that group. The initial branch for root r_ℓ is therefore

$$S = \{r_\ell\}, \qquad F = \{r_1, \ldots, r_{\ell-1}\}.$$

This initial branch is placed on the active branch list. Each time a branch (S, F) is processed, exactly the same sequence of steps is followed:

(1) Compute the total slack $\sum_{i \in S} s_i$ and total value $\sum_{i \in S} V_i$.

(2) Compute the boundary of S relative to F, meaning the counties not in S and not in F that are adjacent to at least one county in S.

(3) If $\sum_{i \in S} s_i \geq 0$, add S to the master list $\mathcal{C}$.

(4) Decide whether the branch is worth expanding further. If the current slack is so

negative that even the remaining positive slack among counties not yet in S and not in F cannot bring total slack back to weakly positive, terminate the branch.

(5) Otherwise, if expansion remains possible, choose one admissible boundary county j according to a fixed rule, such as the lowest-index county in the boundary.

(6) Replace the current branch by two child branches:

$$(S \cup \{j\}, F) \qquad \text{and} \qquad (S, F \cup \{j\}).$$

The first child explores all admissible supersets of S that include j; the second explores all admissible supersets of S that exclude j.

After these children are created, the parent branch (S, F) is removed from the active branch list. The algorithm then moves to the next branch on that list and applies the same six steps again. Once the active branch list is empty, exploration for root r_ℓ is complete, and the algorithm moves on to root $r_{\ell+1}$.

6.1.2 Solve the Weighted Set-Packing Problem

Given $\mathcal{C} = \{C_1, \ldots, C_K\}$ and corresponding weights $\{W_k\}_{k=1}^{K}$, the weighted set-packing problem can be solved by advanced software known as Gurobi. The operations of these algorithms are complex and so not reproduced here, but they produce exact solutions.